

Power for One-Way ANOVA

2026-04-17

Introduction

Power analysis for one-way ANOVA computes the per-group sample size needed to detect a specified pattern of differences among three or more group means with adequate probability. Where the two-sample t -test handles the simple two-group comparison, the one-way ANOVA generalises the comparison to any number of groups, and its power calculation must therefore handle a multi-group effect-size measure rather than a single mean difference. Cohen's f is the standard standardised effect size; it captures the spread of true group means relative to the within-group standard deviation in a single number that maps cleanly to the non-central F distribution under the alternative hypothesis.

Prerequisites

A working understanding of one-way ANOVA, the omnibus F -test, the non-central F distribution, and Cohen's f as the standardised effect size for ANOVA.

Theory

Cohen's f for one-way ANOVA is

$$f = \frac{\sigma_{\text{between}}}{\sigma_{\text{within}}},$$

the SD of the true group means divided by the within-group SD. The relationship to η^2 is $f = \sqrt{\eta^2/(1 - \eta^2)}$. Cohen's conventional benchmarks are 0.10 (small), 0.25 (medium), and 0.40 (large). Under the alternative hypothesis, the omnibus F -statistic follows a non-central F with $(k - 1, N - k)$ degrees of freedom and non-centrality $\lambda = Nf^2$, where N is the total sample size across k groups; power is the tail probability of this non-central F above the critical value.

Assumptions

The design is balanced (equal n per group; modest imbalance is tolerable but reduces efficiency), within-group variances are approximately equal (homogeneity of variance), and residuals are approximately Normal. Welch-corrected variants accommodate variance heterogeneity at the cost of slightly more complex power calculations.

R Implementation

```
library(pwr)

pwr.anova.test(k = 4, f = 0.25, sig.level = 0.05, power = 0.80)

pwr.anova.test(k = 3, f = 0.10, power = 0.80)
pwr.anova.test(k = 3, f = 0.25, power = 0.80)
pwr.anova.test(k = 3, f = 0.40, power = 0.80)
```

```
group_means <- c(50, 55, 58)
pooled_sd   <- 8
grand_mean  <- mean(group_means)
f_est <- sqrt(mean((group_means - grand_mean)^2)) / pooled_sd
f_est
```

Output & Results

`pwr.anova.test()` returns the per-group sample size required to achieve target power at specified f , α , and number of groups. For $k = 4$ groups at medium effect $f = 0.25$, 45 per group (180 total) achieves 80 % power; computing f directly from substantive expected group means and a pooled within-group SD ties the calculation to scientifically defensible assumptions.

Interpretation

A reporting sentence: “With four groups and an anticipated Cohen’s $f = 0.25$ (medium effect by Cohen’s convention; equivalent to expected group means differing by approximately 0.5 within-group SDs), $n = 45$ participants per group (180 total) are required for 80 % power at $\alpha = 0.05$ using a one-way ANOVA. The protocol allocates 50 per group to allow for 10 % attrition, and pre-specified pairwise Tukey HSD comparisons follow if the omnibus is significant.” Always describe f in terms of the underlying group-mean structure.

Practical Tips

- Compute Cohen’s f from substantive expectations about group means and within-group SD wherever possible, rather than relying on Cohen’s medium/large benchmarks; tying the calculation to the actual expected pattern is far more defensible than invoking generic effect-size labels.
- For unequal group variances, power is approximate under the standard formulas; use Welch-adjusted simulation (or `WebPower::wp.kanova()` with explicit variance specification) when heterogeneity is substantial.
- Unbalanced designs have lower power than balanced designs at the same total N ; the loss is small for modest imbalances but grows quickly for ratios more extreme than 2:1. Plan for balance whenever feasible.
- For many groups with a single pre-specified contrast of primary interest (e.g., linear trend across ordered groups, treatment-vs-control contrast), pre-specified contrast tests have higher power than the omnibus F -test and should be used in preference.
- Simulation-based power (`simr`, `Superpower`, `faux`) is more flexible for complex designs — unbalanced cells, heterogeneous variances, non-Normal outcomes, mixed-effects structures — and is increasingly the default approach in modern protocol development.
- For factorial ANOVA (two or more between-subjects factors), the calculation extends naturally with effect sizes for each main effect and interaction; `Superpower::ANOVA_power()` is the standard tool.

R Packages Used

`pwr::pwr.anova.test()` for canonical balanced one-way ANOVA power; `WebPower::wp.kanova()` for an alternative interface and unequal-variance extensions; `Superpower::ANOVA_power()` and `Superpower::ANOVA_exact()` for factorial-ANOVA power simulation; `pwrss::pwrss.f.ancova()` for ANCOVA-power calculations; `simr::powerSim()` for general simulation-based ANOVA power.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.

- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation